Day 006 | Unit 01 | Teacher plan

Positive Rational Numbers: Decimals and Division

Write Zero Digits in Quotients

Standards: 6.NS.B.2

Learning target: I can divide multi-digit whole numbers and check the quotient.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Estimate the quotient first. Divide, multiply, subtract, and bring down one digit at a time. Keep each quotient digit in its place; write zero when needed. Check dividend = divisor × quotient + remainder.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 12

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01 / Learn

Estimate the quotient first. Divide, multiply, subtract, and bring down one digit at a time. Keep each quotient digit in its place; write zero when needed. Check dividend = divisor × quotient + remainder.

Worked example 1

Use long division: 525 ÷ 5. Show the algorithm and check.

1. Estimate how many digits the quotient should have. Work from the leftmost digits of the dividend.

2. 5 ÷ 5: write 1, subtract 5, remainder 0

3. bring down the next digit: 2 ÷ 5: write 0, subtract 0, remainder 2

4. bring down the next digit: 25 ÷ 5: write 5, subtract 25, remainder 0.

5. Check: 5 × 105 = 525.

Answer: 105

Worked example 2

Use long division: 3,175 ÷ 31. Show the algorithm and check.

1. Estimate the quotient first.

2. 31 ÷ 31: write 1, subtract 31, remainder 0

3. bring down the next digit: 7 ÷ 31: write 0, subtract 0, remainder 7

4. bring down the next digit: 75 ÷ 31: write 2, subtract 62, remainder 13.

5. Check: 31 × 102 + 13 = 3175.

Answer: 102 R 13

Day 006 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use long division: 981 ÷ 9. Show the algorithm and check.

02 / Constructed response

Use long division: 1,535 ÷ 5. Show the algorithm and check.

03 / Constructed response

Use long division: 609 ÷ 3. Show the algorithm and check.

04 / Constructed response

Use long division: 8,442 ÷ 21. Show the algorithm and check.

05 / Constructed response

Use long division: 3,360 ÷ 32. Show the algorithm and check.

06 / Constructed response

Use long division: 8,903 ÷ 29. Show the algorithm and check.

07 / Constructed response

Use long division: 3,675 ÷ 36. Show the algorithm and check.

08 / Model and constructed response

Use long division: 6,811 ÷ 33. Show the algorithm and check.

09 / Model and constructed response

Use long division: 2,868 ÷ 14. Show the algorithm and check.

10 / Constructed response

A club has 2447 cards. Each full packet holds 12. How many full packets can it make, and how many cards remain?

Day 006 | Unit 01 | Answer key

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1. 109

9 ÷ 9: write 1, subtract 9, remainder 0; bring down the next digit: 8 ÷ 9: write 0, subtract 0, remainder 8; bring down the next digit: 81 ÷ 9: write 9, subtract 81, remainder 0. Check: 9 × 109 = 981.

6.NS.B.2 | Constructed response

2. 307

15 ÷ 5: write 3, subtract 15, remainder 0; bring down the next digit: 3 ÷ 5: write 0, subtract 0, remainder 3; bring down the next digit: 35 ÷ 5: write 7, subtract 35, remainder 0. Check: 5 × 307 = 1535.

6.NS.B.2 | Constructed response

3. 203

6 ÷ 3: write 2, subtract 6, remainder 0; bring down the next digit: 0 ÷ 3: write 0, subtract 0, remainder 0; bring down the next digit: 9 ÷ 3: write 3, subtract 9, remainder 0. Check: 3 × 203 = 609.

6.NS.B.2 | Constructed response

4. 402

84 ÷ 21: write 4, subtract 84, remainder 0; bring down the next digit: 4 ÷ 21: write 0, subtract 0, remainder 4; bring down the next digit: 42 ÷ 21: write 2, subtract 42, remainder 0. Check: 21 × 402 = 8442.

6.NS.B.2 | Constructed response

5. 105

33 ÷ 32: write 1, subtract 32, remainder 1; bring down the next digit: 16 ÷ 32: write 0, subtract 0, remainder 16; bring down the next digit: 160 ÷ 32: write 5, subtract 160, remainder 0. Check: 32 × 105 = 3360.

6.NS.B.2 | Constructed response

6. 307

89 ÷ 29: write 3, subtract 87, remainder 2; bring down the next digit: 20 ÷ 29: write 0, subtract 0, remainder 20; bring down the next digit: 203 ÷ 29: write 7, subtract 203, remainder 0. Check: 29 × 307 = 8903.

6.NS.B.2 | Constructed response

7. 102 R 3

36 ÷ 36: write 1, subtract 36, remainder 0; bring down the next digit: 7 ÷ 36: write 0, subtract 0, remainder 7; bring down the next digit: 75 ÷ 36: write 2, subtract 72, remainder 3. Check: 36 × 102 + 3 = 3675.

6.NS.B.2 | Constructed response

8. 206 R 13

68 ÷ 33: write 2, subtract 66, remainder 2; bring down the next digit: 21 ÷ 33: write 0, subtract 0, remainder 21; bring down the next digit: 211 ÷ 33: write 6, subtract 198, remainder 13. Check: 33 × 206 + 13 = 6811.

6.NS.B.2 | Model and constructed response

9. 204 R 12

28 ÷ 14: write 2, subtract 28, remainder 0; bring down the next digit: 6 ÷ 14: write 0, subtract 0, remainder 6; bring down the next digit: 68 ÷ 14: write 4, subtract 56, remainder 12. Check: 14 × 204 + 12 = 2868.

6.NS.B.2 | Model and constructed response

10. 203 full packets; 11 cards remain

2447 = 12 × 203 + 11.

6.NS.B.2 | Constructed response